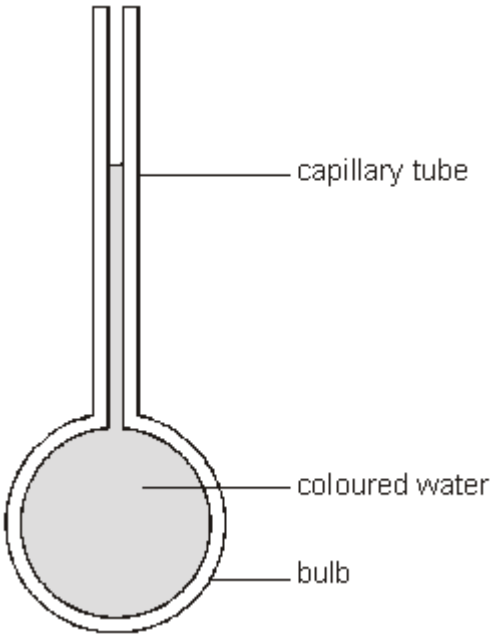
Q1.

Hassan made a simple thermometer by blowing a bulb on the end of a glass capillary tube and filling it with coloured water.



(a) When he warmed the bulb, the water expanded and rose up the tube.
Explain, in terms of the water molecules, why this happened.

.....

.....

.....

2 marks

(b) He wanted to measure the temperature in his freezer, so he left the thermometer in the freezer overnight. The next morning he found that the glass bulb of his thermometer had broken.
Suggest why this had happened.

.....

.....

1 mark

(c) He made a second thermometer which was identical to the first. Why could he not use it to measure the temperature of some hot sugar syrup at 110 °C?

.....

.....

1 mark

(d) Hassan took an identical thermometer at room temperature and placed the bulb in some water at about 60°C. He noticed that the water in the capillary tube dropped slightly before it started to rise.
Tick the box by the correct explanation for his observation.

The volume of water in the thermometer decreased before the water expanded.	☐
The volume of the glass bulb increased before the water expanded.	☐
The volume of the glass bulb decreased before the water expanded.	☐
The volume of the glass bulb decreased before the glass expanded.	☐

1 mark
Maximum 5 marks

Q2.

A pupil has two well-insulated containers of water at 20° C. She takes two identical blocks of aluminium, both at 100° C, and puts one into container A and the other into container B. The temperature of the water rises in each container until it reaches a steady value. Her results are shown in the table.

container	temperature of water in °C at the start	temperature of block in °C at the start	volume of water in cm^3	final temperature of water in °C
A	20	100	500	44
B	20	100	1000	34

(a) Explain why the final temperature of the water is lower in container B.

.....

.....

1 mark

(b) What is the final temperature of each aluminium block?

The block in container A is at .....°C.

The block in container B is at .....°C.

1 mark

(c) Which of the identical aluminium blocks transferred more energy to the water? Explain your reasoning.

.....

.....

1 mark
Maximum 3 marks

Q1.

(a) the water molecules moved faster **or** they had more energy

accept 'they vibrated more'

1 (L7)

the molecules moved further apart

do not accept 'molecules took up more space'
***or** 'they expanded'*

1 (L7)

(b) **a reference to freezing or becoming solid is required for the mark**

any **one** from

- the water expanded as it froze

  accept 'the ice took up more space than the water'

- the water expanded as it became solid

- the pressure in the bulb increased as the water froze

  accept 'it froze in the neck first'

  *do **not** accept 'the water froze and cracked the glass'*
  ***or** 'the water expanded when it was solid'*

1 (L7)

(c) any **one** from

- the water would boil **or** evaporate

- water boils at 100°C

  accept 'the thermometer only goes up to 100°C

1 (L7)

(d) The volume of the glass bulb increased before the water expanded. √

if more than one box is ticked, award no mark

1 (L7)

[5]

Q2.

(a) the energy had to be shared by more particles **or** water

accept 'there was more water to heat up'
***or** 'the heat is spread out more'*
*do **not** accept 'there was more water'*

1 (L7)

(b) A 44

B 34

both answers are required for the mark

1 (L7)

(c) **for the mark to be awarded, the correct block must be identified and the correct reason given**

any **one** from

- the block in B, because its temperature dropped by the larger amount
  both answers are required for the mark
  *do **not** accept 'B, because there was more water to heat up'*

- the block in B, because this block ended up at a lower temperature

1 (L7)

[3]